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Experimental Investigation of Dynamic Pitching Effects on a Delta Wing with Blown Jet

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Active flow control (AFC) devices can be utilized for stall mitigation, enhancing aerodynamic performance, and performing efficient maneuvers by altering the localized flow around different regions of a vehicle. A blown jet was installed along the upper surface of a flat plate delta wing half-span model with a 60° sweep angle in the University of Dayton Water Tunnel (UDWT). This study investigates the use of a blown jet to mitigate the breakdown of the leading-edge vortex. The mass flow rate and the corresponding momentum coefficient were controlled using a closed loop PI controller to alter the lift at various α using the actuator. Forces and moments were recorded to measure the performance. Static and dynamic tests were conducted to measure the change in behavior of vortex breakdown during quasi-steady and unsteady pitching at different moment coefficient. Measurable differences in lift coefficient are observed at a higher angle of attack with AFC. However, the impact of AFC diminishes with an increase in reduced frequency. Preliminary flow visualization revealed a bifurcation of the blown jet exit angle caused by the local pressure gradients at a range of angles of attack depending on the moment coefficient.

Nomenclature

U, V, W	= Velocity in the X, Y, and Z direction	C_μ	= Coefficient of momentum
α	= Angle of attack	k	= Reduced frequency
c_{local}	= Local chord length	LEV	= Leading-edge vortex
C_L	= Coefficient of lift	PIV	= Particle Image Velocimetry
$C_{L,Non-Circ}$	= Coefficient of non-circulatory lift	AFC	= Active flow control
$C_{L,Circ}$	= Coefficient of Circulatory lift	ϕ	= Phase angle
C_M	= Coefficient of moment		

I. Introduction

Moderately swept wings exhibit the formation of a leading-edge vortex (LEV) as a result of separated flow along the leading-edge. The vortex forms on the suction side at high angles of attack and is unique to certain geometries. Relevant geometries include swept planforms such as delta wings, diamond wings, arrow wings, lambda wings, and cranked wings. Typically, the vortex forms with the use of a sharp leading edge to produce the separated flow. The separation and reattachment of flow near the leading-edge produces vorticity that feeds into the LEV along the span. An associated vortex lift is produced as a result providing considerable added lift performance. The LEV is present across a wide range of flight regimes and is sought after as an alternative to maintaining attached flow while

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also producing lift [1]. The magnitude of vortex lift generated increases with vortex strength. Typically, vortex strength is altered by changing the angle of attack and wing geometry to create more vorticity from flow separation. The relationship between geometry, angle of attack, and lift has been explored in literature. Polhamus developed the leading-edge suction analogy to predict lift and drag performance for swept delta wings of various geometries [1]. Lamar has extended the application of the leading-edge suction analogy to low aspect ratio wings [2]. These models show the vortex lift as a function of taper ratio, sweep angle, AR, and Mach number in good agreement with experimental results. However, both prediction models lose accuracy at higher angles of attack where the LEV begins to decay.

Swept wing geometries face disadvantages under certain conditions that can reduce performance and cause instabilities related to the LEV. As the angle of attack increases to higher angles, the vortex core loses strength and begins to expand and break down beginning at the trailing edge. At this point the flow transitions from a jet-like to a wake-like flow [3]. Reverse axial flow is associated with the breakdown found within this wake-like flow [4]. The expansion of the vortex core progresses upstream until the separation meets the apex at which point the LEV then bursts. Increasing the Reynolds Number can delay this breakdown until the stall angle [4][5]. A LEV burst leads to a sudden loss of vortex lift and change in wing dynamics [6]. A non-linear pitching moment is found in response to LEV separation [7] which could pose an issue for traditional control surfaces. Tailless configurations have been shown to be statically unstable in roll from asymmetric vortex breakdown [8][9]. In general, the breakdown of the vortex leads to performance loss as the flow separation fails to maintain a strong vortex core. Influencing the vortex strength through actuation would allow for a way to mitigate the negative effect associated with LEV breakdown. Such methods of actuation to influence the vortex strength include passive and active flow control.

Various flow control methods have successfully altered the LEV dynamics in favorable ways to improve performance [10]. Typically, active methods of flow control involve blown jets. Many variations in jet location and orientation have been explored. Most commonly, jets are located on the upper surface of the wing blowing along the leading edge or located on the leading edge blowing tangential. Alignment of the jet parallel to the leading edge has demonstrated a more coherent LEV [11]. AFC has also demonstrated control authority over pitching and yaw moments with the use of blown jets [7][9][12]. Static roll instability can also be mitigated through AFC [8][9]. These blown jets introduce more momentum locally into the flow to influence the vortex dynamics. However, the amount of C_μ that proves effective can vary by design. Literature shows a non-linear response to C_μ where increasing momentum sees diminishing returns [11][13]. However, the experimental testing of the relationship between C_μ effectiveness and unsteady pitching remain largely unexplored.

The current study aims to use a blown jet actuator to influence the LEV breakdown and resulting C_L in the near stall and post-stall region. Static along with dynamic pitching tests are conducted to investigate the impact of AFC and how it extends in a quasi-steady and unsteady flow. At low angles of attack, the blown jet strengthens the vortex and its effects producing a higher vortex lift. At higher angles of attack, where the vortex would typically break down and burst, the jet adds momentum to the vortex core. This results in a delay in vortex breakdown at the surface of the wing and stabilized response to the vortex. Increasing the reduced frequency led to a decrease in effectiveness from the blown jet actuator.

II. Experimental Setup

A. University of Dayton Water Tunnel

All experiments were performed at the University of Dayton Water Tunnel (UDWT). For a detailed explanation of the experimental setup refer to Mongin and Gunasekaran [14]. The facility is a horizontal free-surface water tunnel with a 6:1 contraction and a 38 cm wide and 46 cm high test section of 152 cm length. The free-surface speed range is 7.5 cm/s to 38 cm/s. The tunnel is equipped with a one degree of freedom motion stage consisting of a H2W Technologies linear motor driven by a Xenus XTL-230-18 motor driver from Copley Controls. This motor is outfitted with a rack and pinon gear that converts its linear motion to a rotational motion to pitch the test article within the tunnel test section. The closed-loop feedback and control for this system is performed with a dSPACE MicroLabBox. Force measurements were performed with an ATI Mini40 force/torque sensor with NET F/T box interfaced with the dSPACE MicroLabBox via the dSPACE Ethernet Blockset. All of the control programming for the dSPACE hardware is done in MATLAB Simulink. The wing model is mounted on the force sensor shown in Figure 1 and the linear actuator is used to change the angle of attack statically and dynamically. It is worth noting that the use of a half span model weakens LEV formation due to the lack of asymmetric vibrations which aid in full-span models [3]. This could potentially mean the half-span model is more susceptible to LEV breakdown.

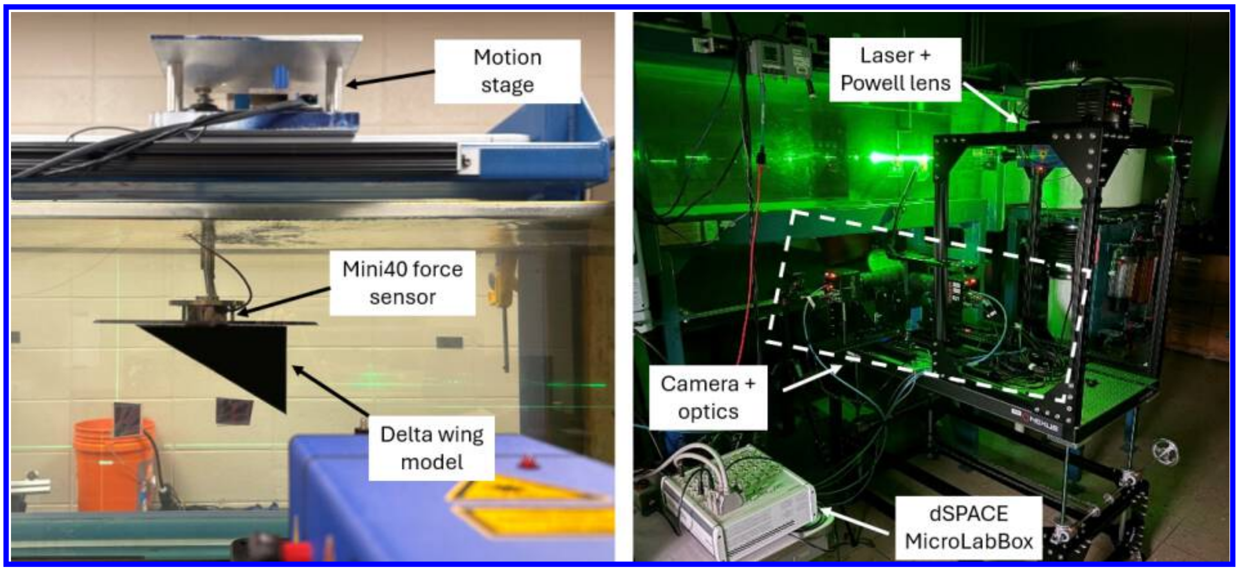

Figure 1. Experimental testing rig at the UDWT for wing model mounted to force sensor (right) and PIV (left)

B. Particle Image Velocimetry Setup

PIV was conducted with a LaVision DaVis 10 collection and processing system. A Photron FastCam Mini WX camera with a 2061 x 2061-pixel sensor and 105mm Nikkor lens was used to image the field of view around the wing. Calibration was performed with a machined dot plate to determine a spatial resolution of 8.66 pixel/mm. The frame rate is adjusted to achieve an 8 to 10-pixel particle displacement between image pairs. A 2.4W, 532nm continuous laser from LaVision was used for experiments in combination with a Powell lens to form the laser sheet. The resulting laser sheet illuminated 60μm polyamide seed particles that were evenly distributed into the flow of the tunnel. Both the camera and laser were mounted to a vertically traversing optical bench capable of adjusting to any spanwise location of the wing while keeping the laser and camera field of view aligned. Figure 1 shows the PIV test setup.

C. Blown Jet Closed Loop Controlled System

A blown jet is emitted from an imbedded nozzle from within the test model to function as an AFC actuator. The nozzle is mounted on the upper surface of the wing model and oriented to blow along the trajectory of the LEV with an intent that the flow from the nozzle accelerates the axial velocity at the core of the vortex to strengthen it. The desired effect of directly adding momentum along the core is to suppress the region of reverse axial flow and therefore maintain a coherent LEV. To supply the actuation port, a pump and subsequent flow meter are used to supply pressure and measure the resulting flow rate.

The AFC system is composed of several components to measure, record, and control flow properties. A SEN-HW06K hall effect flow sensor measures the flow rate from the pump and outputs a corresponding square pulse varying in frequency. An Arduino Uno R3 applies a calibration to the signal and produces a corresponding 0-5V signal output to the MicroLabBox using an MCP 4725 breakout board. The pump is a DC50E continuous circulation pump capable of flow rates up to 26 L/min while maintaining a consistent pressure. The pump motor is controlled with a 5V analog signal to determine the motor speed. Based on the flow sensor measurement feedback, the MicroLabBox throttles the pump to closed-loop control the desired flow rate. The system is designed with modularity in mind to allow for interfacing with either an Arduino, Raspberry pi, or dSPACE MicroLabBox as a controller. A visualization of the closed-loop control is shown in Figure 2. In addition to measuring flow rate, a 5 PSI pressure transducer is included to record data in case of needed flow diagnostics.

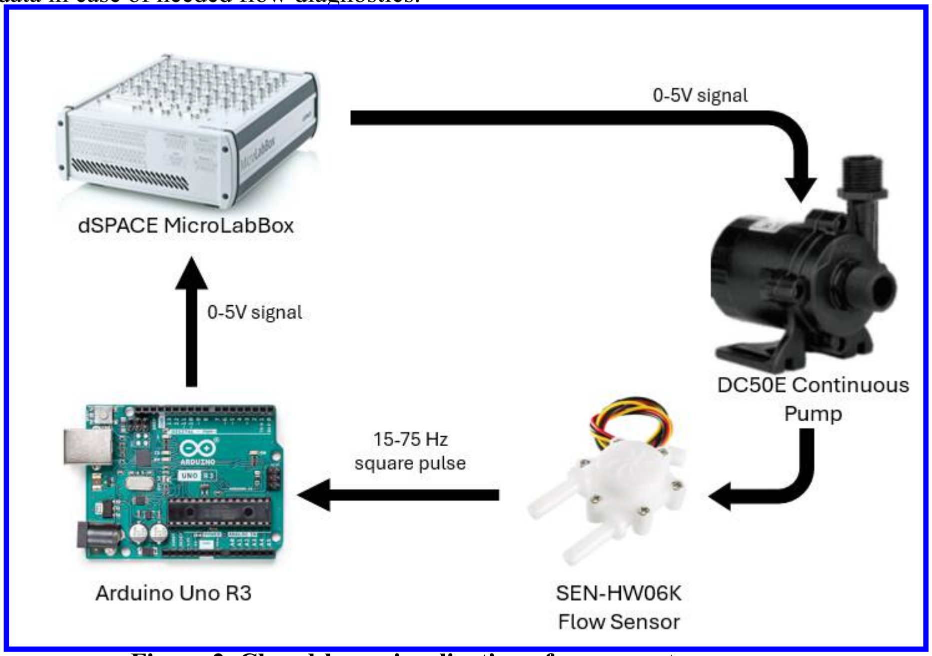

Figure 2. Closed-loop visualization of pump system

To calibrate the flow sensor, first the motor and flow meter were tested in open loop to verify that the manually measured flow rate and sensor output frequency were properly correlated. An Arduino measured the output frequency of the flow meter while the flow rate was measured using a scale and timer. Water was pumped at various fixed rates into a bucket shown in Figure 3. Figure 4 shows the resulting calibration curve applied during experiments.

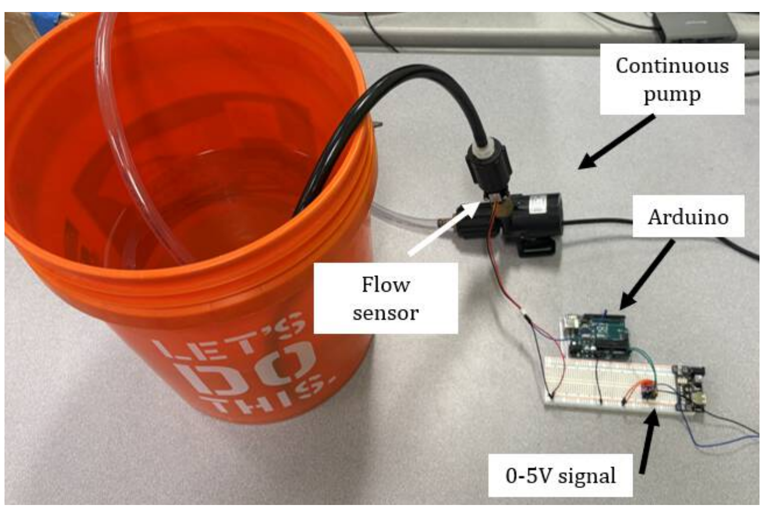

Figure 3. Blown jet system setup for open and closed loop response measurement

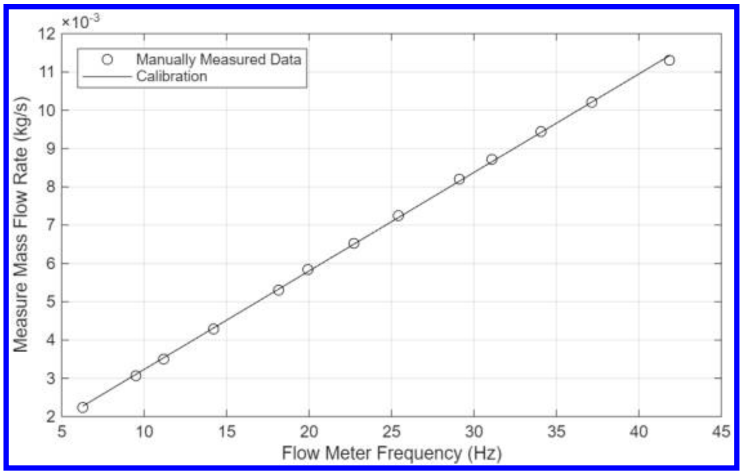

Figure 4. Flow Meter Calibration Curve

D. Test Article Without AFC

To measure baseline performance of a sharp-edge delta wing, a test model with no AFC is investigated to verify the presence of a LEV. The test model shown in Figure 5 is a 6.35 mm thick flat plate delta wing with a 60° sweep angle. The leading-edge geometry is made from a 60° bevel on the lower surface edge leaving a flat upper surface. A puck is integrated at the root of the model for mounting to the force sensor as shown in Figure 1. The half span wing model was fabricated with PLA material using FDM 3D printing. Experimental PIV was tested at various pitch angles as shown in the test matrix in table 1 with a tunnel speed of 0.29 m/s.

Table 1. Test Matrix for PIV

Location	Angle of attack (degrees)	Reynolds Number	Field of View
Root	0, 10, 15, 20, 25, 30	39,000	238 x 238 mm
Mid-span			
Tip			

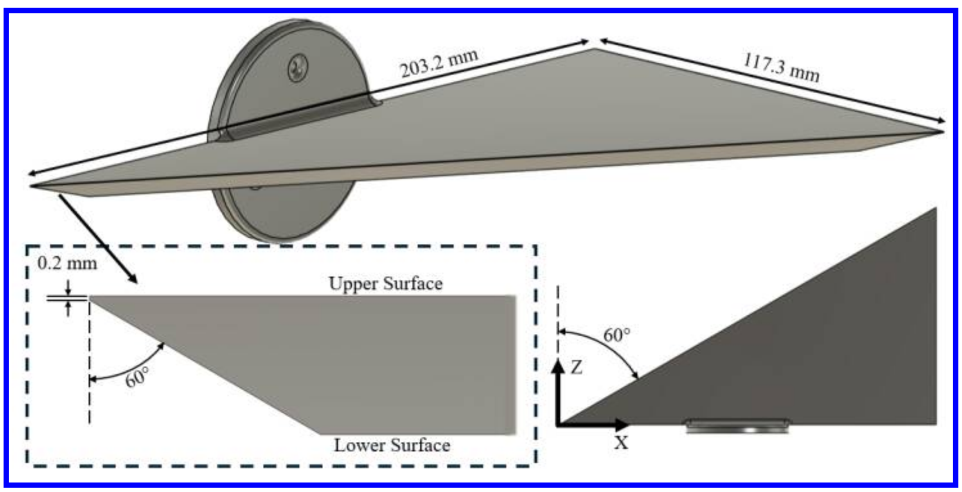

Figure 5. Geometry of baseline delta wing model used in experimental testing

E. Test Article with AFC

The model shown in Figure 5 was modified to include an internal channel to supply the actuator port. Again, the model was FDM printed but now with ABS material to withstand repeated submersion. Flexible tubing from the pump is attached at the model base behind the splitter plate. Figure 6 shows the modified geometry with the internal channel highlighted. The port is positioned at a spanwise distance of $2z/b = 0.245$ and chord location of $x/c_{local} = 0.163$. The direction of the LEV trajectory was determined from initial PIV and determined to have a sweep angle of 51 degrees. Table 2 shows the test matrix for force testing with AFC at different moment coefficients and reduced frequencies. Static force measurements were conducted at various sweep angles. Additionally, the model was dynamically pitched at various reduced frequencies ranging from quasi-steady to unsteady. The peak angle of attack of 35 is chosen to capture near and post stall behavior.

Table 2. Force Measurement Test Matrix

Reduced Frequency	Flow State	C_μ	Angle of attack (deg)	Reynolds Number
0.000	Static	0.000, 0.027, 0.063	-5° to 35°	46,000
0.025	Quasi-steady			
0.050	Unsteady			
0.075	Unsteady			

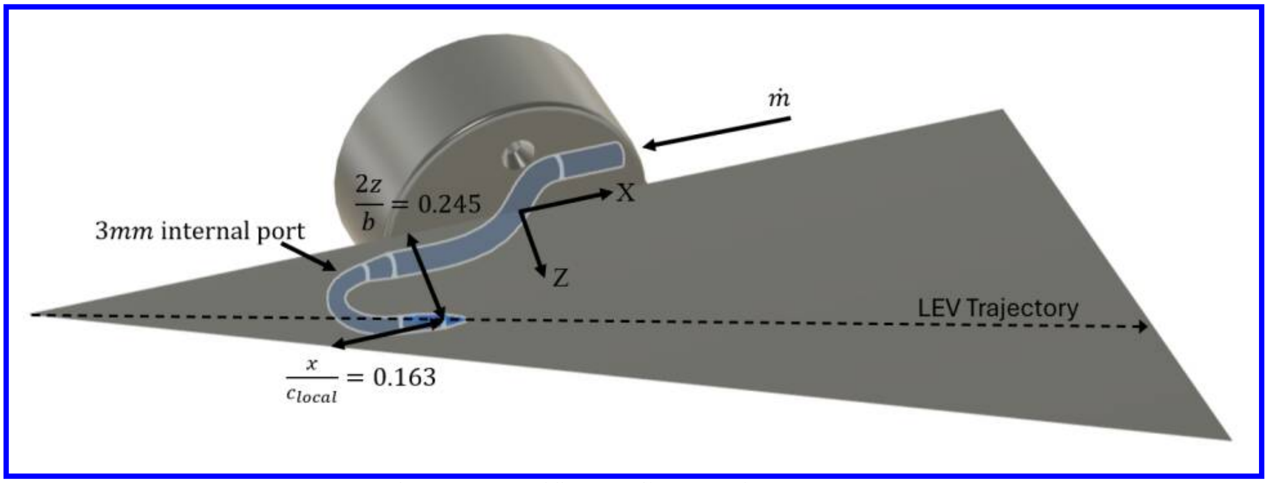

Figure 6. Modified AFC delta wing geometry

Due to the use of AFC during experimental testing, a variety effects from the tubing and flow must be properly understood and tared. First, the tension in the tubing itself holds a non-constant load from changing angle of attack. The resulting pressure from the blown jet also exerts a force on the tube and internal channel of the geometry. Lastly, momentum ejected from the port produces an inertial force along the direction of the jet output. The performance difference strictly due to change in flow field generated by AFC is sought after. For this reason, the inertial and the tube effects are subtracted from the raw measured forces. To measure the effects from the AFC system, forces on the model with tubing attached are measured for all cases in table 2 but with no freestream velocity. Doing so measures the tubing, pressure, and inertial effects produced by the AFC system. For all dynamic cases several cycles were completed and phase averaged together. The beginning of the phase was determined to by when the angle of attack crossed 0 degrees while pitching up.

F. Dynamic Motion Profile

A closed-loop motion stage, labeled in Figure 1, is used to control the pitch angle of the model. Static testing was conducted at discrete points at the specified angles of attack. However, dynamic testing was conducted by pitching the model in a sinusoidal profile ranging across the same angles of attack as the static testing. As such only the frequency of the profile was adjusted to achieve the desired reduced frequency. Figure 7. shows the three different motion profiles used to achieve the specified reduced frequencies.

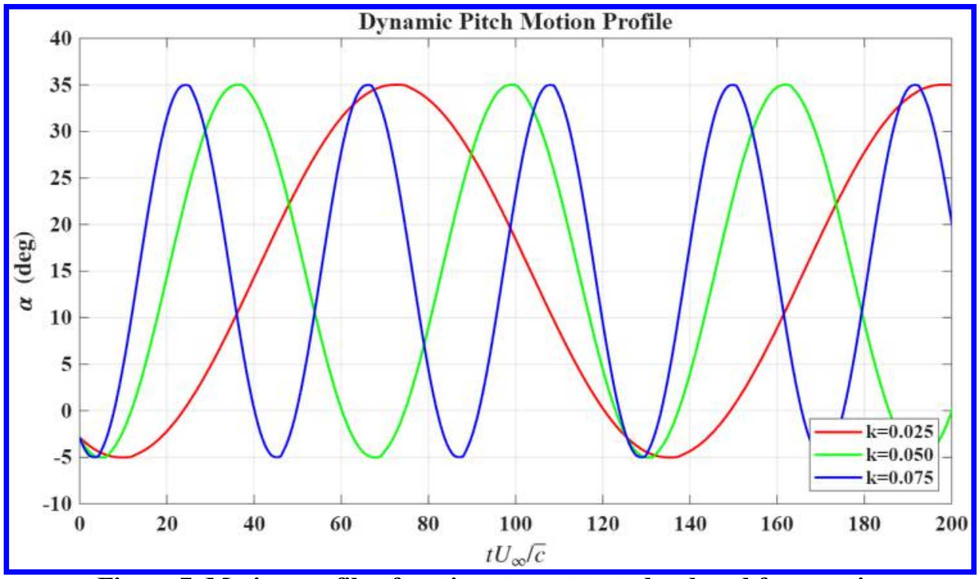

Figure 7. Motion profile of motion stage at tested reduced frequencies

III. Results

A. Flow Visualization

Flow visualization was done on the half-span delta wing model, as shown in Figure 8, to identify the trajectory and breakdown location of the primary leading-edge vortex and to ensure that AFC port was placed along the trajectory of the vortex. A fluorescent dye was injected upstream of the apex using a short tube aligned with the incoming flow. As the dye convected over the half-span delta wing, it rolled up into a coherent LEV that followed a well-defined path from the apex toward the wing tip. Downstream, the dyed vortex core lost coherence and expanded marking the onset of LEV burst. The dashed red curves trace the approximate LEV trajectory and its burst location inferred from these dye streaks. The AFC port was observed to be positioned on the surface directly beneath the pre-burst portion of the LEV path so that the blown jet could add momentum primarily along the vortex core before breakdown occurred.

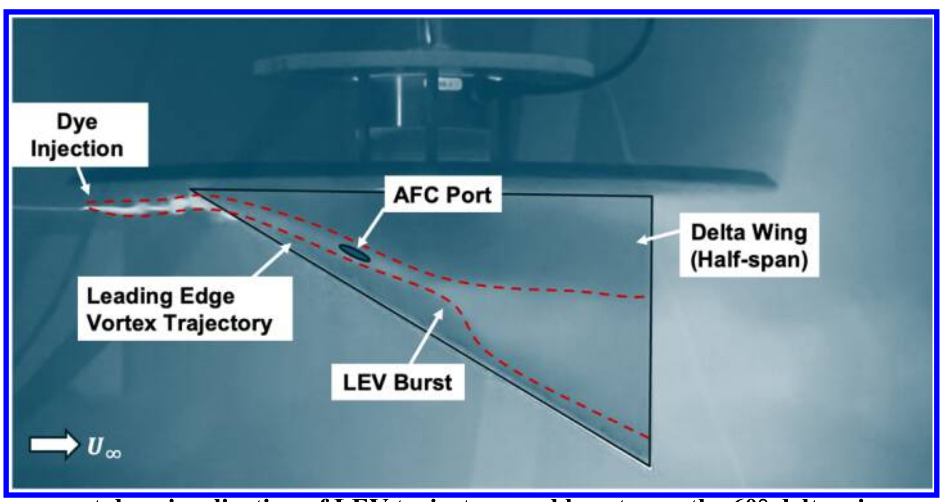

Figure 8. Fluorescent dye visualization of LEV trajectory and burst over the 60° delta wing, used to position the surface AFC port along the vortex core path

B. PIV Results

Streamwise PIV was taken at the root, semi-mid span, and wing tip of the delta wing at a variety of static pitch angles to identify the presence and the location of the vortex as a function of angle of attack. The goal was to identify the angle of attack at which a coherent LEV is observed as well as when it breaks down. Results helped inform the later test matrix in Table 2. The U and V velocity component contours at $\alpha = 10°, 15°,$ and $20°$ are shown in Figure 9. The presence of a strong V-velocity gradient indicates the formation of a LEV at α = 10° and 15°. At 20° no strong V velocity gradient is observed, indicating the breakdown of the LEV at the semi-mid span.

The U velocity of the flow field shown at the same angles of attack indicates that the flow stays attached to the upper surface at the semi-mid span at 10° and 15° angles of attack. In both cases, the flow speeds up in the same region where a Y-velocity gradient appeared in Figure 9. However, at 20 degrees a region of wake-like flow is observed. This again indicates the breakdown of the LEV beyond a 15° static angle of attack. The vorticity contour included in Figure 9 shows indication of a vortex core at 15 degrees. This appears at the same location as the strong V velocity gradient. At 20 degrees there is no longer an observable vortex core.

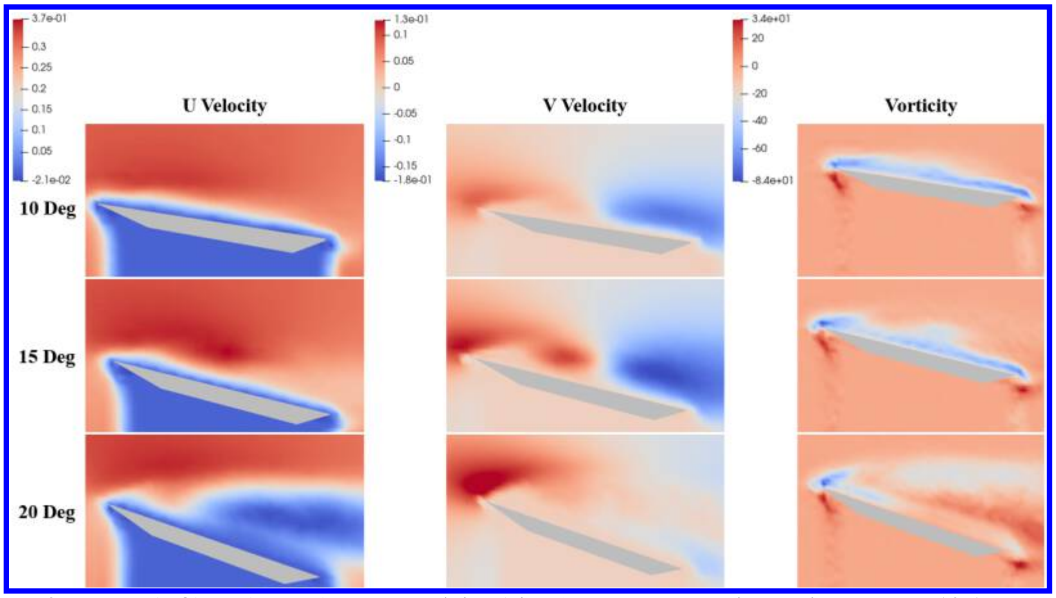

Figure 9. PIV U (left), V (center), and vorticity (right) contours at mid semispan at α=10,15, and 20°

C. Force Results

1. Static Tests

Figure 10a shows the static lift performance of the delta wing with and without AFC. The wing performance with no AFC is considered as a baseline. The lift slope stays mostly constant until 12 degrees where a decrease in slope for the baseline case is observed until stall. Introducing AFC at a fixed $C_\mu = 0.27$ shows a delayed lift slope drop until 17°. Increasing to a $C_\mu = 0.63$ eliminates the lift slope decrease up until stall. Overall, an increase in lift performance is observed at higher angles of attack with the use of AFC. At lower angles of attack up to 12°, there is no performance change associated with AFC. This would suggest that vortex remains coherent until that angle of attack and then bursts at a higher angle of attack as evidenced by the PIV results in Figure 9. It is only after the baseline vortex bursts that the AFC becomes effective. With increase in C_μ, the angle of attack range where vortex remains coherent increases as seen by the increase in lift when compared to no-AFC case. Regardless of AFC, all three cases stall with a peak C_L at 28°. Figure 10b shows the pitching moment performance as measured at the $\bar{c}/4$. The moment produced shows no sensitivity to the blown jet until post-stall. Past this point the baseline sees a sudden drop in pitching moment. However, both AFC cases show no sudden change below 35°.

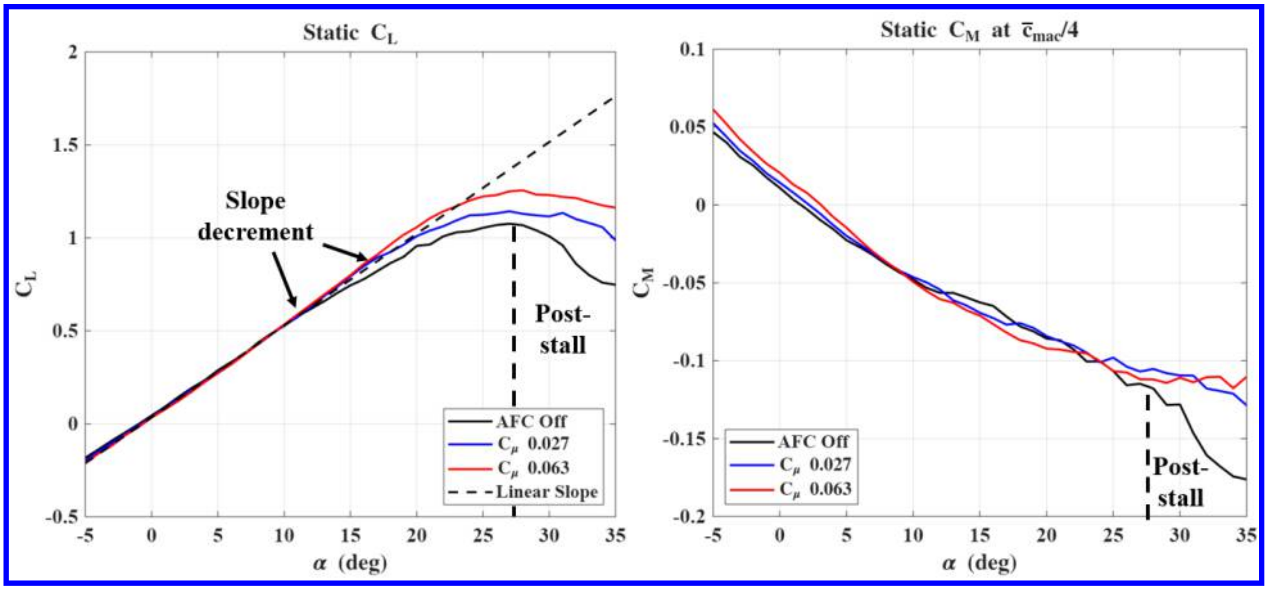

Figure 10. Static a) coefficient of lift and b) coefficient of pitching moment

2. Dynamic Tests

In dynamic pitching cases, the total lift coefficient can be decomposed into two contributions: a circulatory term, $C_{L,Circ}$, and a non-circulatory term, $C_{L,Non-Circ}$ as shown in Equation 1

$$C_{L,Total} = C_{L,Non-Circ} + C_{L,Circ} \tag{1}$$

The circulatory term arises purely from the aerodynamic response of the flow to the prescribed motion, whereas the non-circulatory term accounts for inertial and added-mass effects. Depending on the reduced frequency, these terms can exhibit a phase lag or phase lead relative to the motion profile. In AFC testing, additional non-aerodynamic contributions such as tube forces, pressure, and inertial effects also appear in the forces measured by the sensor.

$$C_{L,Non-Circ} = C_{L,AM+Tube} \tag{2}$$

To determine the efficacy of AFC, it is therefore important to separate the circulatory lift contribution from these other sources in order to analyze how AFC strength modifies the circulatory component of the lift coefficient. The term $C_{L,AM+Tube}$ is determined by pitching the wing with a prescribed motion profile and reduced frequency in still water, in the absence of freestream flow. When AFC is off, $C_{L,AM+Tube}$ includes only the added-mass and tube effects, whereas when AFC is on, the inertial contribution associated with the blown jet is also captured. In the following sections, the variations of $C_{L,\text{AM+Tube}}$ and $C_{L,\text{Circ}}$ are presented and discussed.

2.1 Non-Circulatory Terms

The $C_{L,AM+Tube}$ variation at different reduced frequencies is shown in Figure 11 as a function of angle of attack, with and without the AFC. As the wing moves through the prescribed sinusoidal motion from $-5°$ to 35°, the resulting $C_{L,AM+Tube}$ form a hysteresis loop, with the pitch-up and pitch-down branches annotated in Figure 11. At $k = 0.025$, the C_L magnitude increases with C_μ, a trend observed for all the reduced frequencies tested. However, the size of the hysteresis loop increases with increase in k as expected. This growth in loop area reflects an increasing phase lag or lead between the motion and the non-circulatory response, such that the phase at which the maximum angle of attack occurs is no longer aligned with the phase at which the maximum non-circulatory lift is produced. In addition to the loop growth associated with k, the $C_{L,AM+Tube}$ vs α curves remain consistent for all cases.

Increasing C_μ mostly introduces an upward shift and modest increase in slope. This behavior indicates that the jet-induced tube pressurization and the added mass effects act as an almost linear offset to the non-circulatory lift contribution.

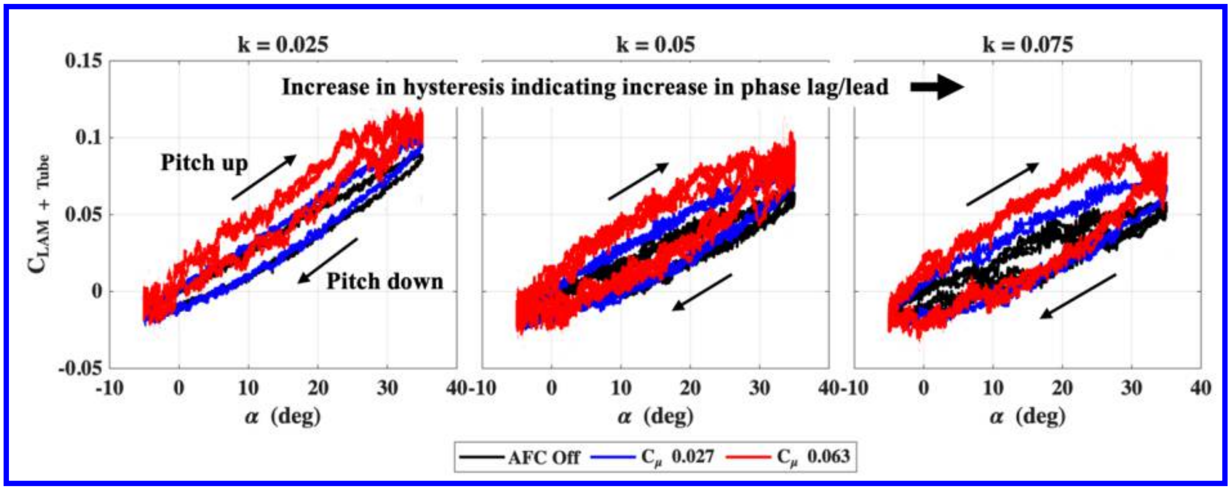

Figure 11. Non-circulatory lift coefficient $C_{L,AM+Tube}$ versus angle of attack for three reduced frequencies, showing increased hysteresis and upward shift with AFC blowing strength.

The same $C_{L,AM+Tube} vs\ \alpha$ results shown in Figure 11 are presented in Figure 12 as functions of convective time and phase ϕ for different reduced frequencies and C_μ. The first column shows the variation over several periods as a function of convective time, while the second column shows the corresponding phase-averaged results. The sinusoidal variation of angle of attack from $-5°$ to $35°$ representing the motion profile is plotted in green. For all reduced-frequency cases, $C_{L,AM+Tube}$ increases with C_μ. The no-AFC case consists primarily of tube effects and added-mass terms. Therefore, the differences between the no-AFC and AFC cases arise primarily from the inertial response to jet blowing and any forces due to pressurization in the tubes. The periodic motion in Figure 12 reveals repeatable trends in each cycle, indicating consistent AFC performance and a stable inertial response over time. This consistency reflects the ability of the closed-loop controller to maintain a nearly constant mass flow rate and, hence, a repeatable momentum injection.

The repeatable behavior over time suggests that the phase-averaged representation in the second column of Figure 12 is a suitable way to characterize the underlying dynamics. The increase in hysteresis-loop size with k, noted in Figure 11, stems from the fact that the phase at which the maximum α occurs differs from the phase at which the maximum $C_{L,AM+Tube}$ occurs. At $k = 0.025$, the phases of the peak α and peak $C_{L,AM+Tube}$ are nearly coincident for all C_μ. As k increases, the phase difference between these peaks grows. The peak $C_{L,AM+Tube}$ occurs before the peak α for each curve, indicating a phase lead that becomes more pronounced with increasing reduced frequency. This behavior is consistent with expectations for dynamic pitching. At $k = 0.075$, a subtle dependence on C_μ is also evident. The phase at which the peak $C_{L,AM+Tube}$ occurs shifts to an earlier ϕ when increasing C_μ. Functionally, this means higher blowing levels at a higher k produce the peak non-circulatory response slightly earlier in the cycle.

In addition to the phase relationships, Figure 12 also highlights how the shape and the amplitude of the non-circulatory response evolve with k and C_μ. At a low k, the $C_{L,AM+Tube}$ waveforms remain aligned in phase with the prescribed sinusoidal motion indicating a predominantly linear inertial response. As k and C_μ increase, the peak-to-peak amplitude grows and a slight flattening of the peak region becomes visible, particularly at the highest blowing level, suggesting an onset of mild nonlinearities in response to the inertial forces by the blown jet and the tube pressurization effects. However, periodic variation indicates a low cycle-to-cycle variability and reinforces that the closed-loop controller delivers a repeatable mass flow rate. This combination of well-defined amplitude, waveform shape, and phase trends provides a robust basis for subtracting $C_{L,AM+Tube}$ from the total lift and isolating the circulatory contribution, which will be discussed in the next section.

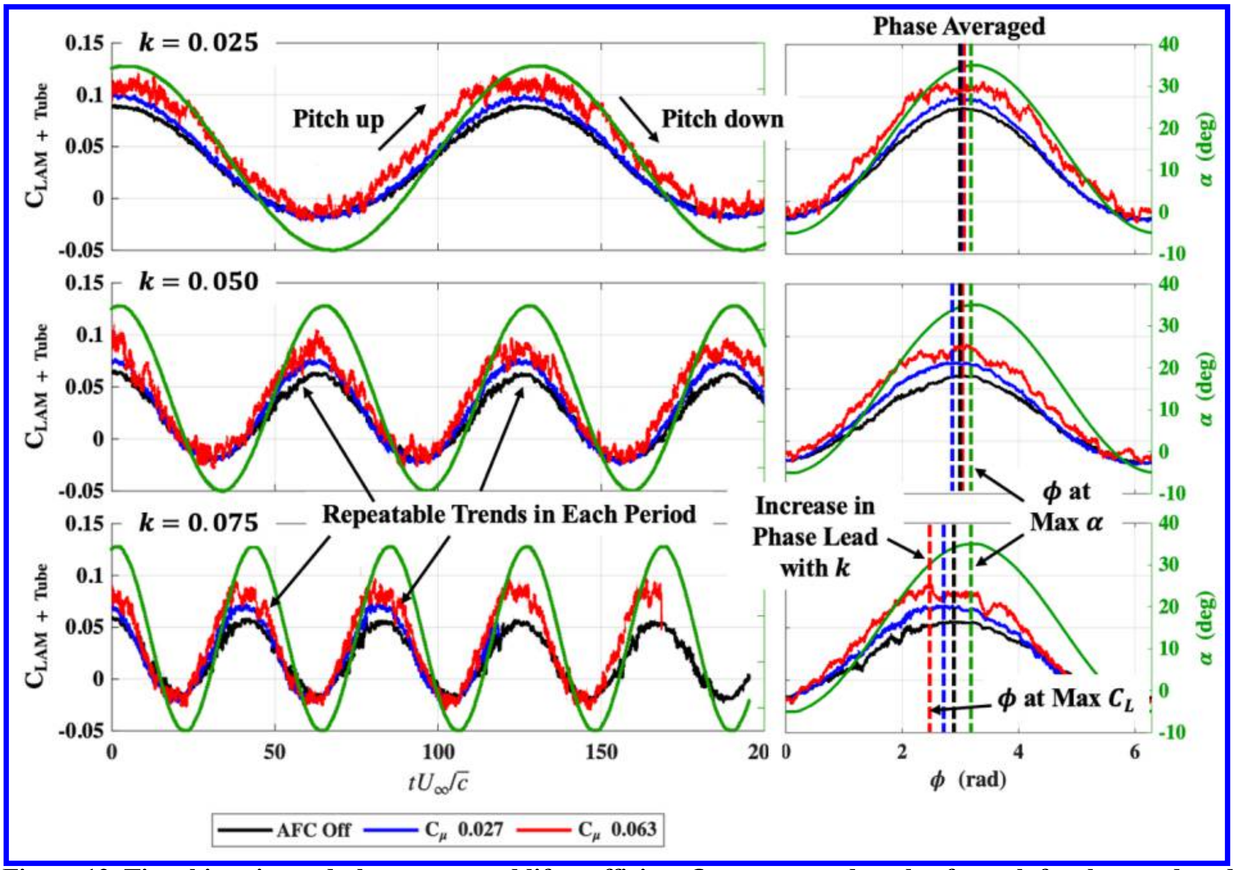

Figure 12. Time histories and phase-averaged lift coefficient $C_{L,AM+Tube}$ and angle of attack for three reduced frequencies, showing increased amplitude and phase lead with AFC blowing strength

2.2 Circulatory Terms

The efficacy of AFC on the aerodynamics of the moderately swept delta wing was evaluated by subtracting $C_{L,\text{AM+Tube}}$ from $C_{L,\text{Total}}$ to isolate the circulatory contribution, $C_{L,\text{Circ}}$. The variation of $C_{L,\text{Circ}}$ with angle of attack is shown in Figure 13 for different reduced frequencies and momentum coefficients. At all reduced frequencies, the wing response remains largely linear with angle of attack, with only a modest hysteresis loop that becomes more apparent at higher k. For each reduced-frequency, increasing C_μ increases the lift coefficient at higher angles of attack and maintains a constant lift-curve slope $\mathrm{d}C_{L,\text{Circ}}/\mathrm{d}\alpha$. These trends are in agreement with the static testing results.

At the quasi-steady reduced frequency, all C_μ cases collapse onto the no-AFC curve up to about $10°$, beyond which a pronounced hysteresis loop forms between $10°$ and $35°$, as highlighted in the magnified $C_{L,\text{Circ}}$ vs α plots in the bottom row of Figure 13. The no-AFC case exhibits the largest loop in this range, with the pitch-down branch lying noticeably below the pitch-up branch, indicating strongly nonlinear, history-dependent behavior. This nonlinearity is likely associated with leading-edge vortex (LEV) burst at high angles of attack followed by a delayed recovery of LEV attachment during the pitch-down portion of the cycle resulting in decreased lift. With increasing C_μ, this lift loss during pitch-down is reduced, leading to a narrow hysteresis loop. In other words, AFC tends to linearize the high-α behavior by mitigating LEV-burst-induced lift degradation and enhancing LEV recovery.

At higher reduced frequencies, this high-α nonlinearity is not observed. The lift recovers during the pitch-down motion even in the no-AFC case, and the pitch-up and pitch-down branches are more closely aligned. For these higher k cases, the hysteresis behavior is similar for AFC and no-AFC, and the dominant effect of AFC is the maintained lift slope across a wider range in angles of attack, with only minor changes in loop shape. This trend suggests that AFC provides its strongest dynamic benefit at low reduced frequencies, where the baseline case suffers the most from LEV recovery. Overall, AFC successfully reduces hysteresis, improves lift recovery, and stabilizes the LEV during slower, large-amplitude maneuvers. At higher reduced frequencies, AFC primarily augments the baseline lift slope while leaving the underlying unsteady characteristics largely unchanged.

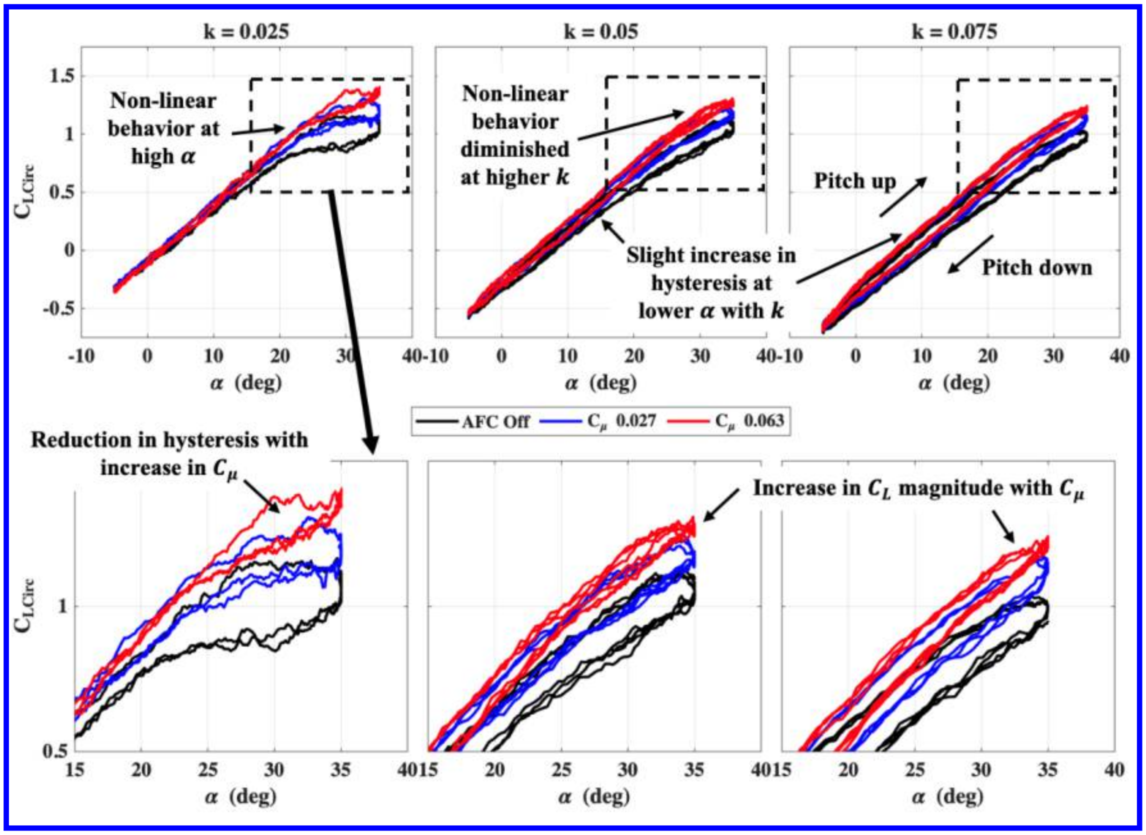

Figure 13. Circulatory lift coefficient $C_{L,\mathrm{Circ}}$ versus angle of attack for three reduced frequencies, showing increased lift and reduced hysteresis with AFC blowing strength, especially at quasi-steady conditions

Similar to Figure 12, the $C_{L,\mathrm{Circ}}$ results from Figure 13 are replotted in Figure 14 as functions of convective time in the first column and phase averaged ϕ in the second column. As expected, the period of the sinusoidal motion is associated with an increasing k. The time histories for both the AFC off and on cases show good repeatability, which indicates that similar dynamics occur in each cycle and that the mass flow rate through the AFC port is maintained consistently. The phase averaged variation of angle of attack and $C_{L,\mathrm{Circ}}$ in the second column therefore gives a compact and representative view of the temporal behavior as the wing pitches at different reduced frequencies.

Consistent with Figure 13, there is little difference between the AFC and no AFC cases at low angles of attack. This overlap follows the trend seen with static testing and shows that AFC mainly affects the flow once the LEV is stronger. Differences become apparent only after the angle of attack exceeds about 20°. For all reduced frequency cases, increasing C_μ increases the circulatory lift at higher angles of attack. The magnitude difference between the no AFC and AFC cases is largest at low reduced frequency and loses impact when k increases. At the quasi-steady reduced frequency ($k = 0.025$), a pronounced flattening of the lift peak appears between about 25° and 35° for both pitch up and pitch down motion until ~20°. This flattening shows strong nonlinearity and indicates a LEV burst and remaining in a burst state during the pitch down portion of the cycle. With increasing C_μ, the angle at which the LEV burst occurs is delayed to a higher α and the range of the flattened region is reduced. The phase averaged curves also become more symmetric about the midpoint of the cycle when AFC is active. In the no

AFC case, the pitch down branch sits noticeably below the pitch up branch, and the AFC reduces this asymmetry and promotes a more complete lift recovery.

At higher reduced frequencies, the flattening of the peak decreases and, by $k = 0.075$, the lift variation is close to sinusoidal. These trends indicate that AFC has its largest influence on LEV dynamics and on the magnitude of C_L at low reduced frequency. The clean, nearly sinusoidal behavior at higher k, highlights that the remaining unsteadiness and nonlinearity at low k are genuinely aerodynamic and are tied to LEV dynamics rather than artifacts of tube inertia. At higher reduced frequencies, AFC mainly provides a global increase in maximum C_L. A slight phase lead is also observed at higher reduced frequencies, which is consistent with the modest increase in hysteresis seen in Figure 13 at $k = 0.075$.

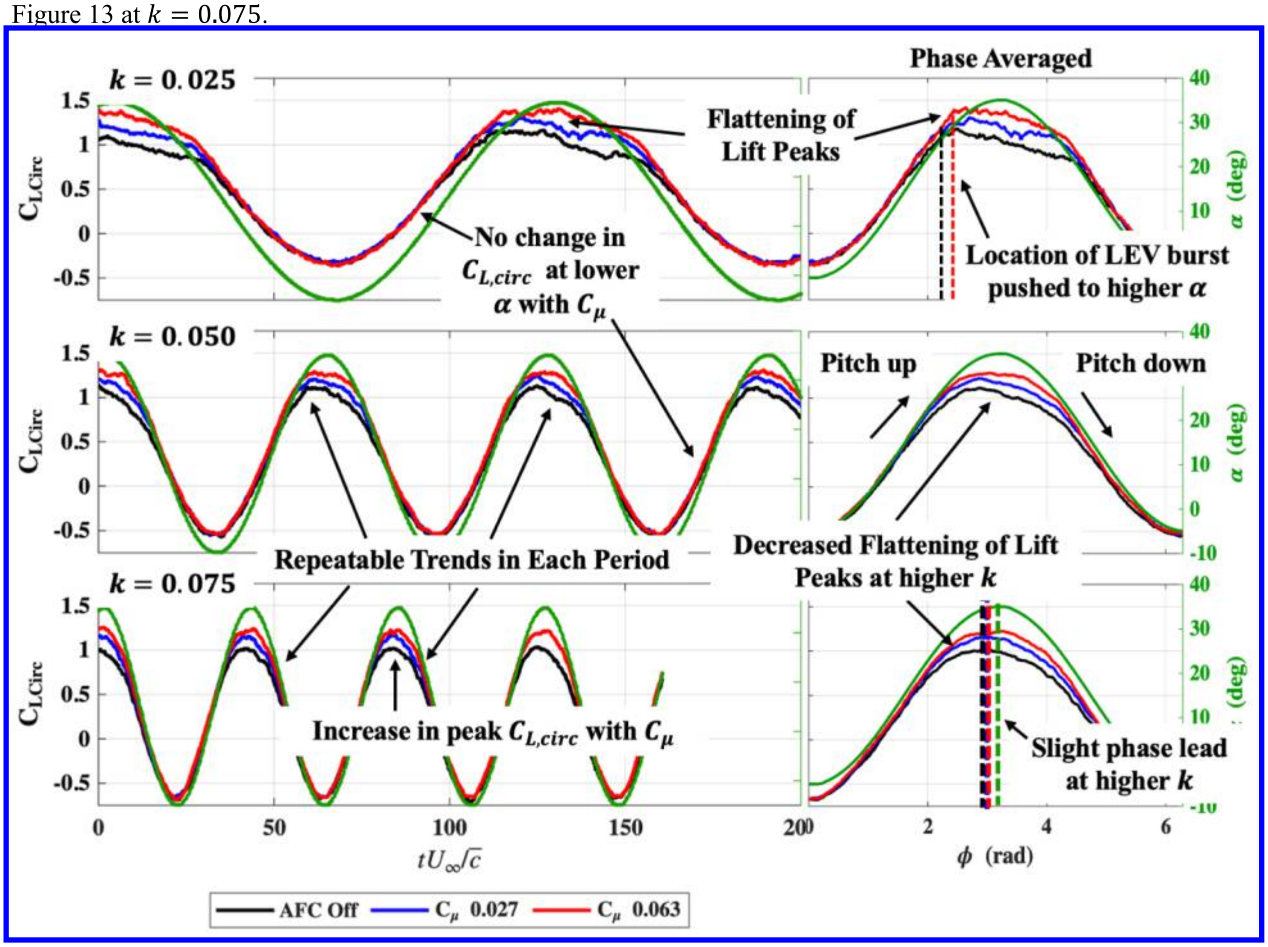

Figure 14. Time histories and phase-averaged circulatory lift coefficient for three reduced frequencies, showing increased peak lift, reduced nonlinearity, and slight phase lead with AFC blowing

2.3 Flow Visualization Results

In order to relate the lift hysteresis to the underlying flow physics, dye was injected through the AFC port to visualize the blown jet and its interaction with the LEV on the delta wing. The top row of Figure 15 shows a low C_μ case and the bottom row shows the higher C_μ case. In each row, the images progress from left to right with increasing angle of attack. At low C_μ and low angle of attack (top left), the jet exits nearly aligned with the freestream and the dye sheet remains close to the local surface streamline. When C_μ is increased (bottom left), the exit trajectory is initially pushed slightly outboard, consistent with the higher momentum flux of the jet relative to the oncoming flow.

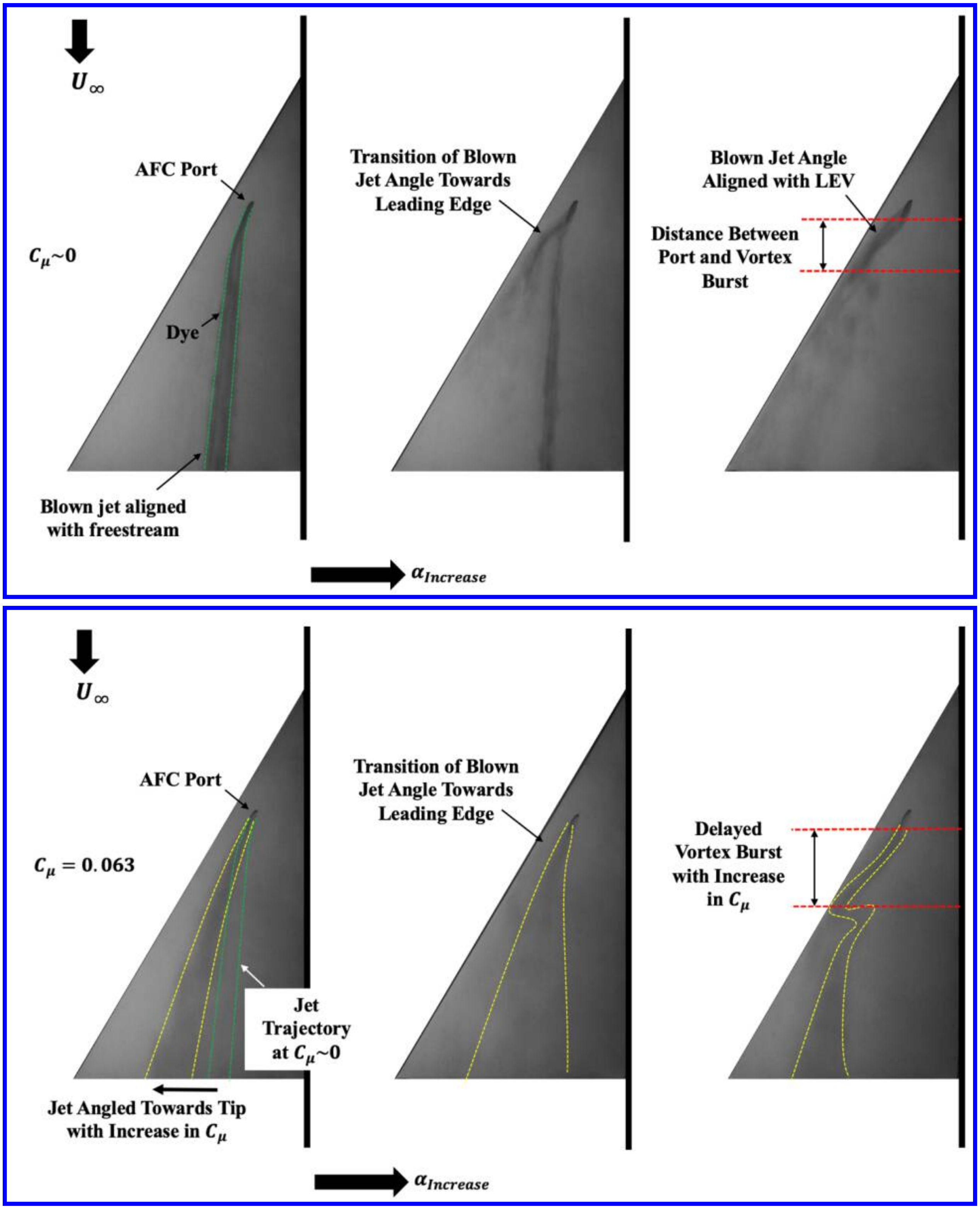

Figure 15. Dye visualizations for low and high C_μ showing jet reorientation toward the leading-edge vortex and delayed vortex burst with increasing C_μ and angle of attack.

As the angle of attack increases, the exit angle of the blown jet begins to pivot. The middle panels show this transition region. For low C_μ the jet reorients toward the leading edge within one or two degrees of angle of attack, whereas at higher C_μ this reorientation occurs more gradually over about four to five degrees. At higher angles of attack, shown in the right panels, the jet ultimately redirects toward the leading edge and aligns with the core of the LEV. This behavior suggests that the low pressure associated with the LEV draws the jet toward the vortex, so the blown jet effectively feeds momentum into the vortex system rather than simply following the freestream direction.

The right-hand images also show the location of vortex burst relative to the AFC port. For the low C_μ case, the LEV breaks down relatively close to the port. Alternatively, for the higher C_μ case the burst location moves farther downstream, and the coherent vortex persists over a longer distance. This downstream shift in LEV burst is consistent with the circulatory lift trends in Figures 13 and 14. At low reduced frequency, the no-AFC case exhibited strong flattening of the lift peak and a large hysteresis loop, which was attributed to LEV burst and incomplete lift recovery during pitch down. With increasing C_μ, the dye visualizations indicate that the vortex remains organized over a longer chordwise extent and bursts later, which explains the higher peak C_L, the reduced flattening of the lift peak, and the more symmetric pitch-up and pitch-down branches.

IV. Conclusions

PIV slices taken at the semi-mid span suggest the presence of a LEV at 10 and 15 degrees followed by its breakdown by 20 degrees. The breakdown of the LEV within this angle of attack range agrees with static force measurements. Baseline static lift results showed a lift slope decrement at 12° due to the breakdown in LEV. Beyond this angle the slope never recovers and further decreases at stall. Utilizing AFC delayed or mitigated the point of lift slope loss to a greater angle of attack up until stall. Additionally, forces near stall reveal a higher peak C_L and post-stall performance with the use of AFC.

Dynamic lift performance was investigated in terms of circulatory and non-circulatory terms. Non-circulatory lift, made up of added mass and tube effects, was subtracted from total measured C_L to solve for circulatory lift. In both terms a phase lead was observed where increasing the reduced frequency and C_μ led to a higher phase lead. Circulatory lift benefits from AFC are realized at higher angles of attack across a range of unsteadiness. Baseline quasi-steady dynamic performance experiences a flattened peak lift curve which was mended with the use of AFC. Higher reduced frequency did not show a flattened peak and therefore did not benefit from AFC in this regard.

Dye flow visualization revealed a bifurcation in jet exit angle associated with angle of attack. The jet stream initially aligns to the freestream at a low angle of attack but transitions to aligning with the leading edge. This is caused by due to the local pressure changes produced by the LEV. Additionally, the indicated LEV burst location is augmented by AFC where a higher C_μ moves the burst location further downstream. Dye flow visualization and force results suggest that benefits in performance are attributed to the movement of LEV further downstream.

Overall, AFC successfully strengthened the LEV to aid in maintaining its coherence in both static and dynamic pitching. AFC produced a sustained lift slope, higher peak C_L, improved LEV pitch down recovery, and reduced hysteresis at high angles of attack. While a higher reduced frequency dampened the benefits seen from LEV recovery and hysteresis reduction, the sustained lift slope and peak C_L still were prominent.

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VI. References

[1] Polhamus, E. C. "Predictions of Vortex-Lift Characteristics by a Leading-Edge Suction Analogy." Journal of Aircraft, Vol. 8, No. 4, 1971, pp. 193–199. https://doi.org/10.2514/3.44254.

[2] Lamar, J. E., "Extension of Leading-Edge-Suction Analogy to Wings with Separated Flow Around the Side Edges at Subsonic Speeds," NASA TR R-428, 1974

[3] Gursul, I., Gordnier, R., and Visbal, M. "Unsteady Aerodynamics of Nonslender Delta Wings." Progress in Aerospace Sciences, Vol. 41, No. 7, 2005, pp. 515–557. https://doi.org/10.1016/j.paerosci.2005.09.002.

[4] Ol, M. V., and Gharib, M. "Leading-Edge Vortex Structure of Nonslender Delta Wings at Low Reynolds Number." AIAA Journal, Vol. 41, No. 1, 2003, pp. 16–26. https://doi.org/10.2514/2.1930.

[5] Kumar, V., Mandal, A. C., and Poddar, K. "An Experimental Investigation on the Aerodynamic Characteristics and Vortex Dynamics of a Flying Wing." The Aeronautical Journal, Vol. 128, No. 1326, 2024, pp. 1681–1705. https://doi.org/10.1017/aer.2023.115.

[6] Veismann, M., Gharib, M., Taubert, L., and Wygnanski, I. "Effect of Leading-Edge Cranks on Stability and Control of Active-Flow-Control-Enabled Tailless Aircraft." AIAA Journal, Vol. 61, No. 9, 2023, pp. 3848–3865. https://doi.org/10.2514/1.j062561.

[7] Campbell, J. F. "Augmentation of Vortex Lift by Spanwise Blowing." Journal of Aircraft, Vol. 13, No. 9, 1976, pp. 727–732. https://doi.org/10.2514/3.58703.

[8] He, X., and Williams, D. R. "Pressure Feedback Control of Aerodynamic Loads on a Delta Wing in Transverse Gusts." AIAA Journal, Vol. 61, No. 4, 2022, pp. 1659–1674. https://doi.org/10.2514/1.j062442.

[9] Williams, D. R., Osteroos, R., Brandt, S., Simon, S., Blue, P., and Lask, A. NATO AVT-350 Task Group: Flight Tests With Active Flow Control Effectors on the D-90-V2 Jet. 2025.

[10] Gupta, S., Kumar, S., and Kumar, R. "Control of Leading-Edge Vortices over Delta Wing Using Flow Control Methods: A Review." Materials Today: Proceedings, Vol. 50, 2021, pp. 2189–2193. https://doi.org/10.1016/j.matpr.2021.09.447.

[11] Naigle, S., Altman, A., and Walker, M. Sensitivity Analysis of LEV Trajectory and Surface Separation to Discrete Jet Design Parameters. 2025.

[12] Kalyankar, H., Wygnanski, I., Higuera Piere Noel, A., Urreiztieta, U., and Taubert, L. NATO AVT-350 Task Group: On Controlling the Path & Structure of a Leading-Edge Vortex by a Single, Small Jet. 2025.

[13] Oshima, E., Gharib, M., and Wygnanski, I. J. Interactions Between the Leading Edge Vortex and Sweeping Jet Actuators on a Simple Swept Wing. 2022.

[14] Mongin, M. P., and Gunasekaran, S., "LIFT regulation using closed-loop feedback and control," AIAA Scitech 2021 Forum, Jan. 2021.